Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 33)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Explain the roles of **three** named groups of soil organisms in maintaining soil fertility.

[6 marks]

0 1 . 2 Explain why a soil pH between 5.5 and 7.0 is best for fertility.

[2 marks]

0 1 . 3 Explain how soil compaction by heavy machinery reduces fertility.

[2 marks]

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0 2 Figure 1 is a flow diagram of the main stages in a sewage treatment works using activated sludge treatment.

Figure 1

Stage	Process
Pre-treatment	Screens and grit traps
Primary treatment	Primary sedimentation
Secondary treatment	Aeration tank then secondary sedimentation
Tertiary treatment	Phosphate stripping / micro-strainers / UV
Sludge treatment	Anaerobic digestion

0 2 . 1 Explain the purpose of screens and grit traps.

[2 marks]

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0 2 . 2 Explain why primary sedimentation is described as the most important process.

[2 marks]

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0 2 . 3 Describe what happens in the aeration tank and explain why the sludge from the secondary sedimentation tank is returned to it.

[3 marks]

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0 2 . 4 Explain why tertiary treatment may be needed before effluent is discharged, and describe **one** tertiary process.

[3 marks]

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0 3 . 1 Describe **four** factors that should be considered when choosing a site for a dam and reservoir.

[4 marks]

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0 3 . 2 Give **one** advantage and **two** disadvantages of an estuary barrage compared with a reservoir on land.

[3 marks]

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Advantage .....

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Disadvantage 1 .....

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Disadvantage 2 .....

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0 3 . 3 Explain why desalination of seawater is only used in some countries.

[2 marks]

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0 3 . 4 Explain how artificial recharge can make aquifer supplies sustainable.

[1 mark]

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0 4 . 1 Complete Table 1 to show a suitable sampling technique for each group of organisms and one limitation of the technique.

[6 marks]

Table 1

Organisms	Technique	One limitation	
Invertebrates active on the ground surface at night			
Invertebrates living in leaf litter			
Invertebrates on tree branches			

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0 4 . 2 Explain why a light trap cannot be used to prove that a moth species is absent from a site.

[2 marks]

0 4 . 3 Explain how a motor-driven suction sampler can give a population density, whereas a pooter cannot.

[2 marks]

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0 5 Table 1 shows typical noise levels of some sounds.

Table 1

Sound	Noise level / dB
Threshold of human hearing (at 1000 Hz)	0
Leaves rustling	20
Library	40
Normal conversation	60
Busy road	80
Chain saw	110
Jet engine at 30 m	140

0 5 . 1 Explain why a logarithmic scale is used to measure sound.

[2 marks]

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0 5 . 2 Use Table 1 to calculate how many times greater the sound intensity of a chain saw is than that of a busy road.

[2 marks]

Answer ..... times

0 5 . 3 A machine produces 85 dB. A second identical machine is switched on next to it. State the new noise level and explain your answer.

[2 marks]

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0 5 . 4 Explain what is meant by the statement that 0 dB does not mean there is no sound.

[1 mark]

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0 5 . 5 State the frequency range of hearing for a typical young person and explain why the dB(A) scale is used for human hearing.

[3 marks]

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0 6 Table 1 compares two types of satellite used to monitor the Earth.

Table 1

Feature	Low Earth Orbit (polar) satellite	Geostationary satellite
Altitude	About 800 km	36 000 km above the equator
Orbit period	About 1.5 hours	24 hours (fixed position)
Coverage	Whole Earth surveyed in about 15 days	Same third of the Earth continuously
Detail	High resolution	Lower resolution

0 6 . 1 Use Table 1 to explain why LEO satellites are used to monitor changes in ice cover and vegetation, whereas geostationary satellites are used to monitor weather.

[3 marks]

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0 6 . 2 Give **three** climate-related factors, other than ice cover, that are measured by satellite sensors.

[3 marks]

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0 6 . 3 Explain how a computer climate model can be tested for reliability before its predictions are used.

[2 marks]

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0 6 . 4 Explain why representative data are essential for climate modelling.

[2 marks]

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0 7 . 1 Which of the following best describes the purpose of Critical Group Monitoring? Tick one box.

[1 mark]

A To detect health damage in the population most affected by a pollutant ☒

B To predict the route a pollutant will take through the environment ☐

C To assess exposure of the people most at risk so that everyone else can be assumed to be sa ☒

D To measure the concentration of a pollutant at the point of release ☐

0 7 . 2 Explain why CPA can reduce the cost of an environmental monitoring programme.

[2 marks]

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0 7 . 3 A new chemical has properties similar to PCBs (persistent, liposoluble chlorinated organic compounds). Explain how knowledge of PCBs helps decisions about the new chemical.

[2 marks]

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0 8 Table 1 shows the concentration of cadmium in rice and in the kidneys of people living in two villages in Japan. Village A is downstream of a zinc mine.

Table 1

Village	Cadmium in rice / mg kg^{-1}	Cadmium in kidney of residents / mg kg^{-1}	Cases of bone disease (itai-itai) per 1000 people
A (downstream of mine)	1.2	180	34
B (control)	0.05	22	0

0 8 . 1 Calculate how many times more concentrated cadmium is in the kidneys of village A residents than in their rice.

[1 mark]

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Answer [ ] times

0 8 . 2 Explain why cadmium and zinc pollution often occur together.

[1 mark]

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0 8 . 3 Use Table 1 and your knowledge of the properties of cadmium to explain the high rate of bone disease in village A.

[4 marks]

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0 8 . 4 Suggest **two** ways cadmium pollution from the mine could be controlled.

[2 marks]

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0 8 . 5 Cadmium is used in cadmium-telluride photovoltaic panels. Suggest why this new use may cause pollution in the future.

[2 marks]

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0 9 . 1 Describe polymetallic (manganese) nodules and explain why they are not yet exploited on a large scale.

[3 marks]

0 9 . 2 Explain the environmental impacts of mining polymetallic nodules.

[3 marks]

0 9 . 3 Explain why rare earth metals are a supply concern and describe one new extraction technique.

[4 marks]

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1 0 . 1 Discuss how the features of energy resources determine their suitability for different uses and their environmental sustainability.

[9 marks]

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Extra space

1 0 . 2 State **one** reason why nuclear power stations can be located where transport infrastructure is poor.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the extent to which the study of ozone depletion illustrates how scientific investigation can lead to the successful solution of an environmental problem.

[25 marks]

OR

1 1 . 2 Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS